

Training report for StarDist 2D model (model_4_400epoch_no_aug)

Date: 2025-10-20

Training time: 5.0hour(s) 24.0min(s) 25sec(s)

Information for your materials and method:

The StarDist 2D model was trained for 300 epochs on 800 paired image patches (image dimensions: (1000, 1000), patch size: (512,512)) with a batch size of 90 and a mae loss function, using the StarDist 2D ZeroCostDL4Mic notebook (v 1.20.5) (von Chamier & Laine et al., 2021). The model was retrained from a pretrained model. Key python packages used include tensorflow (v 2.19.0), csbdeep (v 0.8.1), cuda (v 12.5.82 Build cuda_12.5.r12.5/compiler.34385749_0). The training was accelerated using a GPU.

Augmentation: No augmentation was used for training.

Parameters

The following parameters were used for training:

Parameter	Value
number_of_epochs	300
patch_size	512x512
batch_size	90
number_of_steps	8
percentage_validation	10
n_rays	32
grid_parameter	2
initial_learning_rate	0.0002

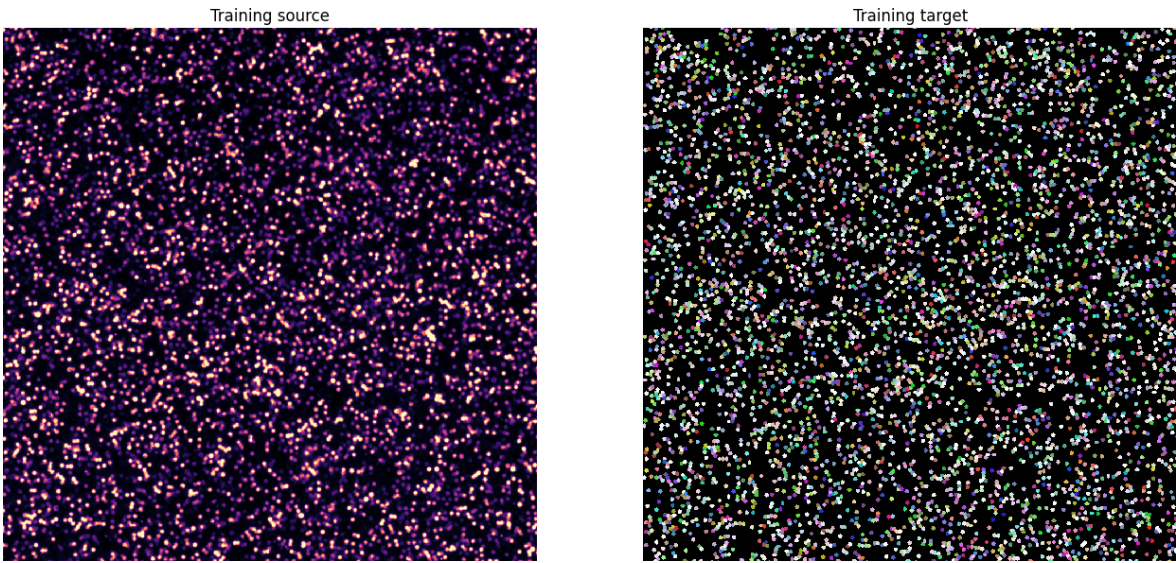
Training Dataset

Training_source: /content/data/training_data_4/KC24 2512/images

Training_target: /content/data/training_data_4/KC24 2512/masks

Model Path: /content/model_4_400epoch_no_aug/model_4_400epoch_no_aug

Example Training pair



References:

- ZeroCostDL4Mic: von Chamier, Lucas & Laine, Romain, et al. "Democratising deep learning for microscopy with ZeroCostDL4Mic." Nature Communications (2021).
- StarDist 2D: Schmidt, Uwe, et al. "Cell detection with star-convex polygons." International Conference on Medical Image Computing and Computer-Assisted Intervention. Springer, Cham, 2018.

Important:

Remember to perform the quality control step on all newly trained models
Please consider depositing your training dataset on Zenodo